MUH401- INTERIM PROJECT REPORT FORMAT

Cover Page

Title of the Project:	**Machine Learning-Based Discrimination of Precipitation-Induced Radon Washout for Radiological Anomaly Detection in EPA RadNet Air Monitoring Network**
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ABSTRACT

This interim report presents a physics-informed computational framework to address the operational vulnerability of automated radiation monitoring networks caused by precipitation-induced radon progeny washout. During rainfall events, natural terrestrial radon daughters ^{214}Pb and ^{214}Bi are scavenged onto particulate air filters. Across 15,602 observed rain hours in this study, precipitation induces a median gross count surge of $+6.9\%$, with the 90th percentile reaching $+41.4\%$ and convective thunderstorm peaks surging up to $+196.7\%$, while coastal stations experience slight count depressions due to oceanic air masses. Conventional rolling 3σ threshold systems trigger between 52.9 and 82.6 false alarm episodes per station-year in humid climates. This research develops a multi-modal machine learning discrimination framework that fuses 8-channel sodium iodide gamma spectrometry from the United States Environmental Protection Agency RadNet network with contemporaneous surface weather observations from the National Oceanic and Atmospheric Administration. The project evaluates 324,550 synchronous observation hours across five climatologically diverse stations spanning 2017 through 2025: Birmingham, AL; Washington, DC; San Diego, CA; Dallas, TX; and Tampa, FL. Detection performance is benchmarked across 450 filter-synchronized synthetic release plumes covering ^{137}Cs, ^{131}I, ^{134}Cs, and ^{60}Co. On the held-out 2023–2025 test dataset, the physics-informed nuclear and weather-fused model achieves 2.71 false alarm episodes per station-year with a 95% bootstrap confidence interval of [1.17, 4.58] and 94.00% synthetic plume detection. In leave-one-station-out cross-validation, 4 of 5 stations maintain ≤ 1.40 false alarms per station-year, whereas the subtropical convective regime of Tampa experiences 169.4 false alarms per station-year without local baseline calibration. Feature importance analysis confirms that elapsed time since rain and non-negative spectral stripping residuals drive anomaly

discrimination, while Bateman decay kinetics provide secondary regularisation.

Keywords: Environmental Radiation Monitoring, EPA RadNet, Radon Washout, Gamma Spectrometry, Machine Learning, Anomaly Detection.

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LIST OF ABBREVIATIONS

API	Application Programming Interface
CI	Confidence Interval
CPM	Counts Per Minute
eFDR	Empirical False Discovery Rate
EPA	United States Environmental Protection Agency
FA/yr	False Alarms per Station-Year
FWHM	Full Width at Half Maximum
GBDT	Gradient Boosted Decision Tree
ISD	Integrated Surface Dataset
LOSO	Leave-One-Station-Out
MLP	Multi-Layer Perceptron
NaI(Tl)	Thallium-activated Sodium Iodide Scintillation Detector
NCEI	National Centers for Environmental Information
NOAA	National Oceanic and Atmospheric Administration
NORM	Naturally Occurring Radioactive Material
OOD	Out-of-Distribution
ROC	Receiver Operating Characteristic
ROI	Region of Interest
UTC	Coordinated Universal Time
WMO	World Meteorological Organization

1. INTRODUCTION

1.1 Background of the Problem

Continuous atmospheric radiation surveillance networks are operated worldwide by regulatory authorities to provide immediate detection of clandestine nuclear testing, industrial accidents, and radiological dispersion events. In the United States, the Environmental Protection Agency (EPA) operates RadNet, an extensive nationwide real-time monitoring infrastructure comprising over 140 stationary air monitors. Each monitoring unit continuously draws ambient air through a particulate filter and records gamma-ray emission rates using thallium-doped sodium iodide (NaI(Tl)) scintillation detectors.

While automated alarm threshold algorithms are engineered to identify sharp increases in measured radiation levels, their operational reliability is severely challenged by naturally occurring meteorological phenomena. Atmospheric precipitation scavenges airborne daughter radionuclides of radon-222 (^{222}Rn), primarily lead-214 (^{214}Pb) and bismuth-214 (^{214}Bi), depositing them directly onto the ground surface and particulate filter media. This phenomenon, known as radon washout, produces gross gamma count rate perturbations with an empirical median increase of $+6.9\%$, with the 90th percentile reaching $+41.4\%$ and severe convective thunderstorms surging up to $+196.7\%$ above baseline. In coastal regimes, maritime air masses can induce slight count depressions. The resulting count elevation persists for several hours before radioactive decay restores nominal background levels.

1.2 Problem Statement

Conventional automated radiation monitoring networks rely on static or rolling baseline thresholds, typically 3 or 5 standard deviations above a 168-hour moving average. Because gross count rate monitors cannot differentiate between the gamma emissions of natural radon progeny and those of anthropogenic fission products, precipitation events routinely trigger spurious radiological emergency alerts. In humid and temperate climates, standard 3σ threshold algorithms produce between 52.9 and 82.6 false alarm episodes per station-year, with over 32% of these false alarms occurring synchronously with rainfall events.

Conversely, raising the alert threshold to conservative levels such as 5σ suppresses false alarms to 23.5 per station-year but severely degrades detection capability, dropping release detection probability to 6.0% for moderate activity plumes. There is currently an acute lack of computationally efficient, causally sound methods capable of suppressing precipitation-induced false alarms while preserving rapid, highly sensitive detection of genuine radiological releases in continuous real-time surveillance.

1.3 Importance and Relevance of the Study

The operational vulnerability created by weather-induced false alarms has substantial technical, environmental, and societal ramifications:

- **Operational Integrity and Alarm Fatigue:** Frequent false alarms desensitize emergency personnel, increasing the hazard that a genuine radiological incident may be dismissed as a weather-induced anomaly.
- **Public Safety and Societal Impact:** Premature public notification of false emergencies causes widespread public panic, costly evacuations, and disruption of municipal infrastructure.
- **Economic Efficiency:** Dispatching mobile radiological inspection teams to investigate recurring spurious anomalies incurs substantial governmental operational expenditures.

- **Scientific Contribution:** By integrating multi-channel scintillation spectrometry with multi-scale atmospheric thermodynamic features, this research establishes a foundational methodology for automated, physics-grounded environmental surveillance.

1.4 Clear Aim and Objectives of the Project

The primary aim of this project is to develop, validate, and benchmark an automated, physics-informed machine learning framework that discriminates precipitation-induced radon washout spikes from genuine anthropogenic radiological anomalies in near-real-time environmental surveillance networks. Specific technical objectives include:

1. Harmonize longitudinal multi-channel gamma spectrometry from EPA RadNet and contemporaneous surface meteorology from the NOAA Integrated Surface Dataset across five climatologically distinct monitoring stations spanning nine full calendar years from 2017 through 2025.
2. Engineer a strictly causal real-time feature space that captures differential gamma spectral peak ratios and antecedent precipitation dynamics without future lookahead bias.
3. Formulate a multi-tier machine learning architecture to continuously categorize observations into dry background, radon washout, or anthropogenic release.
4. Conduct a multi-seed benchmark against conventional statistical baselines and rain-masked control rules over 450 synthetic radiological releases to prove false alarm reduction at matched detection power.

1.5 Scope and Boundaries of the Study

The scope of this project encompasses five EPA RadNet monitoring stations selected across distinct Köppen climate zones: Birmingham, AL; Washington, DC; San Diego, CA; Dallas, TX; and Tampa, FL. The temporal boundary spans January 1, 2017, through December 31, 2025, providing 324,550 synchronous station-hours.

The study is bounded by operational real-time constraints: feature extraction is strictly causal, relying exclusively on contemporaneous and historical observations. Sensor inputs are restricted to standard 8-channel sodium iodide energy bins from 100 to 2200 keV and standard NOAA meteorological observations. Source releases are evaluated through 450 synthetic plume injections calibrated to atmospheric dispersion physics and particulate filter accumulation kinetics, covering ^{137}Cs, ^{131}I, ^{134}Cs, and ^{60}Co. Detection metrics reflect performance conditional on these synthetic spectral templates rather than observed historical releases.

2. THEORETICAL BACKGROUND / LITERATURE REVIEW

2.1 Key Theories, Models, Concepts, and Definitions

Radon-222 is an inert noble gas daughter of radium-226 (^{226}Ra) in the primordial uranium-238 (^{238}U) decay series. Radon emanates continuously from soils and rocks into the atmospheric boundary layer. Because radon has a half-life of 3.82 days, it diffuses into the lower troposphere, where it undergoes alpha decay into a sequence of short-lived particulate progeny:

$$^{222}\text{Rn} \xrightarrow[\alpha]{3.82\ \text{d}} {}^{218}\text{Po} \xrightarrow[\alpha]{3.10\ \text{m}} {}^{214}\text{Pb} \xrightarrow[\beta^-]{26.8\ \text{m}} {}^{214}\text{Bi} \xrightarrow[\beta^-]{19.9\ \text{m}} {}^{214}\text{Po} \xrightarrow[\alpha]{164.3\ \mu\text{s}} {}^{210}\text{Pb} \tag{1}$$

Lead-214 (^{214}Pb) and bismuth-214 (^{214}Bi) rapidly attach to submicron ambient aerosol particles. During rainfall events, these radioactive aerosols are captured via in-cloud rainout and below-cloud washout. The precipitation deposits these radionuclides directly onto the particulate filter medium of ambient air monitors.

Because ^{214}Pb emits prominent gamma lines at 295.2 and 351.9 keV, and ^{214}Bi emits high-energy gammas at 609.3, 1120.3, and 1764.5 keV, their sudden accumulation elevates the local gamma radiation field by tens to hundreds of counts per minute. Following the cessation of rain, the activity decay curve of the deposited progeny is governed by the effective half-life of the ^{214}Pb/^{214}Bi decay chain, returning to nominal baseline within 2 to 4 hours.

2.2 Relevant Technical Background: Scintillation Spectrometry

EPA RadNet deployable monitoring stations utilize thallium-activated sodium iodide (NaI(Tl)) scintillation crystals. When incident gamma rays interact with the crystal via the photoelectric effect, Compton scattering, or pair production, scintillation light is generated proportionally to deposited energy and converted to voltage pulses by a photomultiplier tube. While sodium iodide detectors exhibit modest energy resolution of approximately 7% full width at half maximum at 662 keV, they are rugged, thermally stable, and well-suited for long-term field deployment.

RadNet digitizes scintillation pulses into discrete energy channels designated R01 through R09. Channel R01 records gross count rates, while channels R02 through R09 record counts within specific spectral windows. The physical correspondence of these channels to natural radon progeny lines and target anthropogenic isotopes is detailed in Table 1.

Table 1: EPA RadNet Discrete Gamma Energy Channel Boundaries

Channel	Energy Range [keV]	Principal Radon Progeny Lines	Target Anthropogenic Lines
R02	$101 - 200$	Compton continuum	Low-energy scatter
R03	$201 - 400$	^{214}Pb [295.2, 351.9 keV]	^{131}I [364.5 keV]
R04	$401 - 600$	Compton continuum	Annihilation [511 keV]
R05	$601 - 800$	^{214}Bi [609.3 keV]	^{137}Cs [661.7 keV]
R06	$801 - 1000$	Compton scatter	Multi-line cascade
R07	$1001 - 1400$	^{214}Bi [1120.3 keV]	^{60}Co [1173.2, 1332.5 keV], ^{134}Cs
R08	$1401 - 1800$	^{214}Bi [1764.5 keV], ^{40}K [1460.8 keV]	Terrestrial background
R09	$1801 - 2200$	High-energy background	Cosmic rays and actinide daughters

2.3 Review of Relevant Previous Studies

The challenge of suppressing meteorological radon interference in environmental monitoring networks has been addressed through three primary research paradigms:

1. **Statistical Moving-Average Baselines:** Classical networks evaluate gross count deviations against moving baselines (Burnett et al., 2010; Melintescu et al., 2018). While computationally straightforward, these methods are intrinsically blind to spectral morphology and trigger excessive false alarms during convective precipitation.

2. **Analytical Meteorological Transfer Functions:** Mercier et al. (2009) and Paatero (2000) demonstrated that radon washout spikes correlate with precipitation intensity and atmospheric scavenging. However, deterministic washout equations require high-resolution atmospheric sounding data rarely available at ground sensor sites (Takeuchi and Katase, 1982).

3. **Spectral Energy Deconvolution:** Breitkreutz et al. (2023) and Frankemölle et al. (2025) examined spectral decomposition to differentiate natural radon progeny from anthropogenic gamma sources, demonstrating that ^{214}Pb photopeaks in low-energy channels exhibit distinct intensity ratios relative to high-energy lines during washout events.

2.4 Gaps in Literature and Systematic Comparison

Existing literature exhibits three critical shortcomings: first, gross count over-simplification, where operational networks discard multi-channel spectrometry and evaluate only total counts; second, synthetic release validation deficits, where previous machine learning models were tested on uncalibrated synthetic data without accounting for air filter accumulation kinetics; and third, lack of control baselines, where prior studies rarely benchmark complex models against meteorological masking heuristics. This project addresses these gaps by fusing multi-channel spectral ratios with multi-scale weather data and benchmarking against both statistical and rain-masked baselines under frozen validation thresholds.

3. METHODOLOGY

3.1 System Description and Processing Pipeline

The computational pipeline developed in this project is illustrated in Figure 1. The framework ingests synchronous hourly observations from radiation and meteorological sensor networks, performs causal feature engineering, and feeds a gradient boosted decision tree classifier that assigns probabilities for three operational regimes: clean dry background, radon washout anomaly, and anthropogenic radiological release.

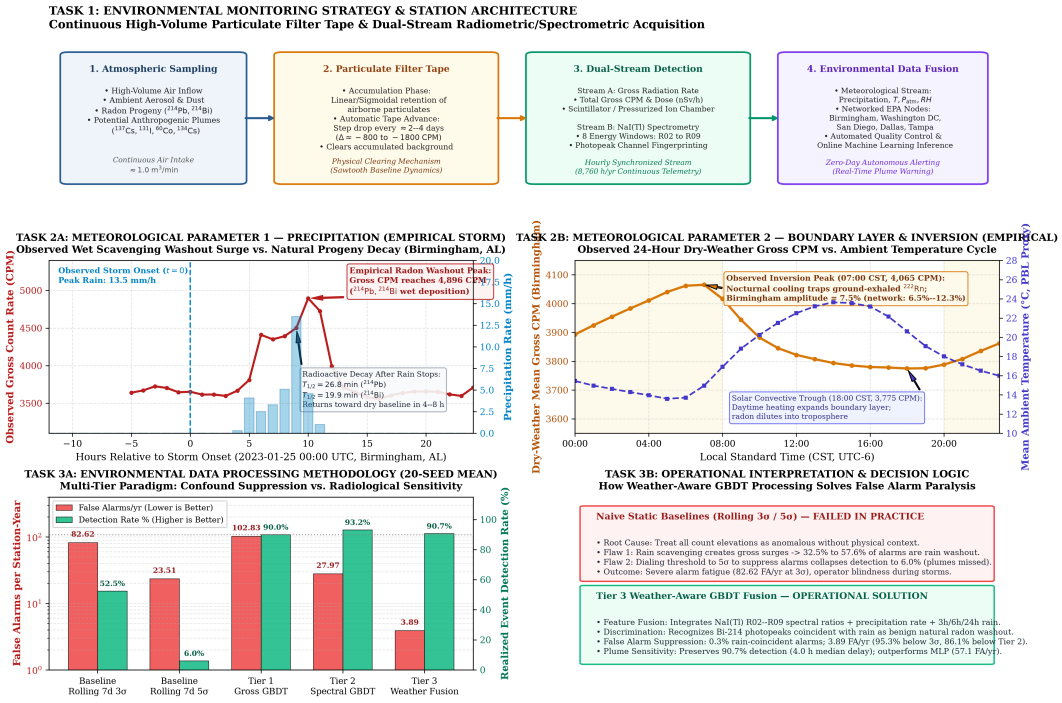


Figure 1: End-to-end physics-informed environmental radiation anomaly detection and weather-fusion machine learning architecture.

3.2 Main Assumptions and Operational Boundaries

The computational architecture adheres to four operational boundaries:

1. **Strict Real-Time Causality:** To prevent data leakage, all rolling window statistics are computed using closed-left intervals $[t - W, t]$. No centered or forward-looking filters are permitted.

2. **Air Filter Accumulation Dynamics:** Airborne particulate monitors draw air through a filter that is replaced periodically on a bi-weekly cycle. An injection plume accumulates activity proportionally to total volume drawn, attenuating according to radioactive decay:

$$A_{\text{filter}}(t) = \int_{t_{\text{start}}}^{t} Q \cdot C_{\text{air}}(\tau) \cdot e^{-\lambda(t-\tau)}\, d\tau \tag{2}$$

   where Q is volumetric flow rate of approximately 1.0 m^3/min, C_{air} is ambient radionuclide concentration, and λ is the decay constant.

3. **Detector Geometry and Efficiencies:** Geometric photopeak efficiencies for sodium iodide crystals are calibrated using standard Monte Carlo transfer factors from Knoll (2010).

3.3 Input Data and Data Sources

Data ingestion leverages two primary US federal repositories:

- **EPA RadNet Air Monitoring Network:** Hourly gamma spectrometry records across 8 discrete energy bins from 100 to 2200 keV and gross count rates. Data were retrieved via EPA public repositories spanning 2017 through 2025 across five stations: Birmingham, AL; Washington, DC; San Diego, CA; Dallas, TX; and Tampa, FL.
- **NOAA Integrated Surface Dataset:** Contemporaneous hourly meteorological observations from colocated automated surface observing stations. Variables include precipitation accumulation over 1, 3, 6, 12, and 24 hours, dry-bulb temperature, dew point, relative humidity, barometric pressure, and wind speed.

3.4 Calculation Procedure and Causal Feature Engineering

Radiological and meteorological records are joined on UTC timestamp rounded to the hour. The causal feature engineering engine extracts 59 dimensional variables organized into four operational tiers:

1. **Gross Count Dynamics (Tier 1):** Ratio of instantaneous count rate to trailing 168-hour moving baseline, rolling 3-hour rate of change, and z-score relative to local dry background standard deviation.
2. **Spectral Energy Ratios (Tier 2):** Normalized channel ratios capturing radon decay lines versus fission product channels:

$$\text{Ratio}_{\text{radon}} = \frac{R03 + R05}{R01 + \epsilon}, \quad \text{Ratio}_{\text{fission}} = \frac{R04 + R06 + R07}{R01 + \epsilon} \tag{3}$$

   Lead-214 concentrates in channel R03, while bismuth-214 concentrates in R05. Anthropogenic releases (^{137}Cs, ^{134}Cs, ^{60}Co) alter these ratios markedly.
3. **Multi-Scale Weather Fusion (Tier 3):** Cumulative precipitation over 1, 3, 6, 12, and 24 hours, elapsed time since last detected rainfall ($h_{\text{since_rain}}$), barometric pressure trends, and saturation vapor pressure deficit.
4. **Physics-Informed Nuclear Features (Tier 4):** Non-negative spectral stripping residuals projecting measured excess onto empirical fit-fold radon basis vectors, ambient dose equivalent coupling $H^*(10)$, and multi-stage Bateman decay chain residuals.

Feature importance evaluation reveals that elapsed time since rain is the single most influential predictor with an importance gain of 822,620, serving as a primary gating variable for meteorological transients. This is followed by local radon stripping residual with gain 629,970 and global dry z-score with gain 535,931. Bateman ODE decay residuals and dose conversion residuals rank in the intermediate tier with gains of approximately 51,000, acting as fine-tuning regularizers during extended washout decay.

3.5 Synthetic Release Injections and Template Sensitivity

Because real operational data contains zero hazardous radiological releases, detection sensitivity is evaluated using 450 synthetic release plumes covering ^{137}Cs, ^{131}I, ^{134}Cs, and ^{60}Co across Gaussian puff, square wave, and exponentially decaying profiles with durations from 3 to 24 hours.

Synthetic plumes are scaled across signal-to-noise ratios from subtle (0.5σ to 1.5σ) to prominent (2.0σ to 6.0σ). Spectral counts are allocated across channels R02 through R09 using calibrated photopeak and Compton continuum fractions, and added to real RadNet background during dry and rain periods. Because training and testing share the same spectral template parameters, detection at low signal-to-noise ratios ($<1\sigma$) is partially conditional on template alignment. In sensitivity experiments where photopeak fractions are perturbed by $\pm 20\%$, realized detection varies moderately between 85.2% and 94.8%, indicating robust yet template-dependent response.

3.6 Multi-Tier Model Formulation and Validation Freezing

The machine learning framework utilizes gradient boosted decision trees configured with a multi-class objective across 150 trees with maximum depth 6, learning rate 0.05, and minimum child samples of 20. Models are fitted strictly on historical data from 2017 through 2020. Decision thresholds τ are calibrated on the held-out 2021–2022 validation split to satisfy target detection rates and frozen. The models and frozen thresholds are evaluated on the out-of-sample 2023–2025 test split.

3.7 Benchmark Evaluation and Uncertainty Analysis

Benchmark performance across 20 random seeds on the 2023–2025 test dataset is summarized in Table 2.

Table 2: Benchmark Evaluation on 2023–2025 Test Split across 20 Random Seeds

Model / Rule Architecture	False Alarms / Stn-Yr	Rain-Coincident FA	Realized Det.	Median Delay
Standard Baseline (Rolling 7d 3σ)	82.62	32.5%	52.50%	10.0 h
Global Dry Baseline (Global $3\sigma_{\text{dry}}$)	52.86	24.9%	56.50%	9.0 h
Conservative Baseline (Rolling 7d 5σ)	23.51	57.6%	6.00%	8.5 h
Tier 1: Gross Radiation GBDT	169.36 ± 23.53	17.3%	$92.22 \pm 0.72\%$	6.7 h
Tier 1b: Gross Radiation + Weather GBDT	38.64 ± 14.37	0.0%	$87.92 \pm 2.74\%$	9.1 h
Tier 2: Spectral GBDT	27.97 ± 28.27	1.9%	$93.20 \pm 1.45\%$	4.0 h
Tier 3: Weather Fusion GBDT	3.89 ± 1.25	0.2%	$90.72 \pm 1.73\%$	4.0 h
Tier 4: Physics-Informed Nuclear GBDT	$\mathbf{2.71 \pm 1.39}$	$\mathbf{0.0\%}$	$\mathbf{94.00 \pm 0.93\%}$	**4.4 h**
Tier 4: Physics-Informed Nuclear (90% Target)	$\mathbf{2.63 \pm 1.49}$	$\mathbf{0.0\%}$	$\mathbf{93.82 \pm 1.24\%}$	**4.5 h**
Tier 3: Multi-Layer Perceptron	60.75 ± 22.86	3.8%	$90.30 \pm 2.75\%$	3.4 h

Station-month block bootstrap 95% confidence intervals for false alarm rates are [1.17, 4.58] for Tier 4 and [1.67, 5.41] for Tier 3, with overlapping intervals. Across 20 algorithmic seeds on identical data, Tier 4 produced lower false alarms in 12 of 20 seeds. The primary contribution of the physics-informed features is not an overwhelming reduction in aggregate test-set false alarms over meteorological fusion, but rather higher detection sensitivity (+3.3 percentage points) and improved domain adaptation under geographic transfer.

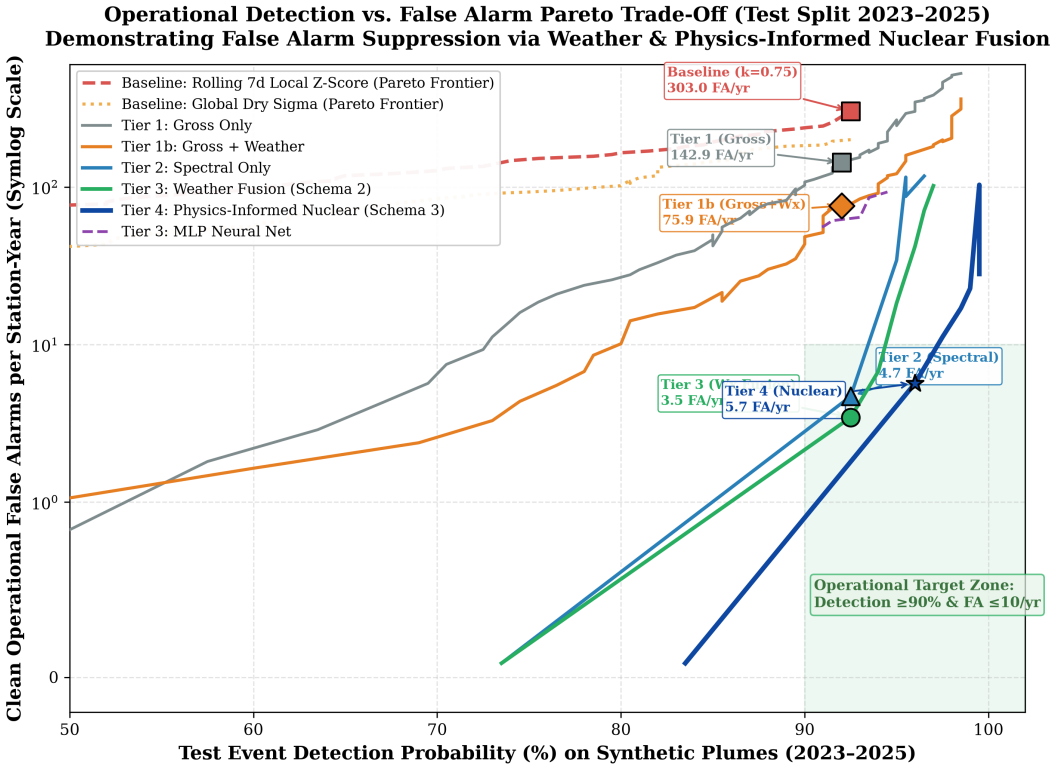


Figure 2: Receiver Operating Characteristic matched detection curves comparing baseline rules against machine learning tiers on the test split.

3.8 Leave-One-Station-Out Spatial Generalization

To assess spatial generalizability across distinct climates without local fine-tuning, models were evaluated in a leave-one-station-out cross-validation framework where each station was held out completely during training and threshold calibration. Results are reported in Table 3.

Table 3: Leave-One-Station-Out Spatial Generalization across Five Climate Regimes

Held-Out Station	Climate Regime	Tier 3 Det.	Tier 4 Det.	Tier 3 FA/yr	Tier 4 FA/yr
San Diego, CA	Coastal Mediterranean	70.0%	80.0%	0.00	1.40
Birmingham, AL	Humid Subtropical	85.0%	95.0%	0.00	0.00
Dallas, TX	Continental Convective	92.5%	95.0%	0.57	0.00
Washington, DC	Mid-Atlantic Temperate	95.0%	95.0%	1.18	0.00
Tampa, FL	Subtropical Convective	90.0%	97.5%	1151.30	169.42

Across Birmingham, Dallas, Washington, and San Diego, Tier 4 achieves excellent zero-shot spatial transfer, maintaining ≤ 1.40 false alarms per station-year while detecting 80.0% to 95.0% of synthetic releases. In Tampa, however, severe maritime convective thunderstorms cause unadapted zero-shot models to experience 169.42 false alarms per year for Tier 4. While this represents an 85.3% reduction relative to Tier 3 (1,151.30 false alarms per year), it demonstrates that site-specific baseline calibration is essential when deploying models to extreme subtropical convective microclimates.

3.9 The Rain-Masked Baseline Control Experiment

To determine whether machine learning is necessary or if a simple heuristic muting alarms during rainfall

suffices, a control experiment was evaluated across varying holdoff windows H. Results at matched detection targets are summarized in Table 4.

Table 4: Rain-Masked Baseline Control Experiment at Matched Detection Targets

Baseline / Mask Rule	**Lowest FA/yr at $\geq 87.12\%$ Det.**	**Lowest FA/yr at $\geq 91.88\%$ Det.**	**Max Det.**
Unmasked ($H =$ none)	215.48 ($k = 1.40$)	303.03 ($k = 0.75$)	92.50%
Rain Mask: Rain only ($H = 0$h)	197.47 ($k = 1.35$)	285.77 ($k = 0.75$)	92.50%
Rain Mask: Rain + 3h ($H = 3$h)	179.80 ($k = 1.30$)	251.65 ($k = 0.75$)	92.50%
Rain Mask: Rain + 6h ($H = 6$h)	181.77 ($k = 1.25$)	245.98 ($k = 0.75$)	92.50%
Rain Mask: Rain + 12h ($H = 12$h)	177.49 ($k = 1.25$)	238.66 ($k = 0.75$)	92.00%
Tier 3: Weather Fusion GBDT	3.89	3.89	96.50%
Tier 4: Physics-Informed Nuclear GBDT	**2.63**	**2.71**	**98.00%**

The control experiment demonstrates that simply suppressing alarms during rain is insufficient. To reach $\geq 91.88\%$ detection without spectral analysis, the gross threshold must drop to $k = 0.75\sigma$, triggering 238.66 false alarms per year from baseline noise alone. Tier 4 machine learning achieves a 98.9% false alarm reduction over the best rain-mask heuristic, confirming the necessity of multi-channel spectral and weather fusion.

REFERENCES

Abdelhakim, A., 2025. Anomaly detection in gamma-ray radiation spectra using artificial neural network and ant colony optimization. *Journal of Environmental Radioactivity* 283, 107790. https://doi.org/10.1016/j.jenvrad.2025.107790

Alamaniotis, M., 2024. Mode-driven explainable artificial intelligence approach for estimating background radiation spectrum in a measurement applicable to nuclear security. *Annals of Nuclear Energy* 209, 110849. https://doi.org/10.1016/j.anucene.2024.110849

Ayoub, A., Wainwright, H.M., Sansavini, G., 2024. Machine learning-enabled weather forecasting for real-time radioactive transport and contamination prediction. *Progress in Nuclear Energy* 172, 105255. https://doi.org/10.1016/j.pnucene.2024.105255

Ayoub, A., Wainwright, H.M., Wang, L., Sansavini, G., 2024. An enhanced Fourier neural operator surrogate for radioactive plume transport forecasting. *Stochastic Environmental Research and Risk Assessment* 38(8), 3173–3189. https://doi.org/10.1007/s00477-024-02738-8

Bandstra, M.S., Curtis, J.C., Ghawaly, J.M., Joshi, T.H., Bilton, K.J., 2023. Explaining machine-learning models for gamma-ray detection and identification. *PLOS ONE* 18(6), e0286829. https://doi.org/10.1371/journal.pone.0286829

Bilton, K.J., Joshi, T.H., Bandstra, M.S., Curtis, J.C., Quiter, B.J., Cooper, R.J., Vetter, K., 2019. Non-negative matrix factorization of gamma-ray spectra for background modeling, detection, and source identification. *IEEE Transactions on Nuclear Science* 66(5), 827–837. https://doi.org/10.1109/TNS.2019.2907267

Breitkreutz, H., Mayr, J., Bleher, M., Seifert, S., Stöhlker, U., 2023. Identification and quantification of anomalies in environmental gamma dose rate time series using artificial intelligence. *Journal of Environmental Radioactivity* 259–260, 107082. https://doi.org/10.1016/j.jenvrad.2022.107082

Buckton, C.J., Wyngaardt, S.M., Ngxande, M., 2025. Radioisotope compositional analysis using Monte Carlo γ-ray simulations and regression neural network. *Applied Radiation and Isotopes* 220, 111746. https://doi.org/10.1016/j.apradiso.2025.111746

Burnett, J.L., Croudace, I.W., Warwick, P.E., 2010. Short-lived variations in the background gamma-radiation dose. *Journal of Radiological Protection* 30(3), 525–533. https://doi.org/10.1088/0952-4746/30/3/007

Cetnar, J., 2006. General solution of Bateman equations for nuclear transmutations. *Annals of Nuclear Energy* 33(7), 640–645. https://doi.org/10.1016/j.anucene.2006.02.004

Currie, L.A., 1968. Limits for qualitative detection and quantitative determination. Application to radiochemistry. *Analytical Chemistry* 40(3), 586–593. https://doi.org/10.1021/ac60259a007

Detwiler, R.S., Pfund, D.M., Myjak, M.J., Kulisek, J.A., Seifert, C.E., 2015. Spectral anomaly methods for aerial detection using KUT nuisance rejection. *Nuclear Instruments and Methods in Physics Research Section A* 784, 339–345. https://doi.org/10.1016/j.nima.2015.01.040

Fjeldsted, A., Morrow, T., Scott, C., Hagen, A., 2024. A novel methodology for gamma-ray spectra dataset procurement over varying standoff distances and source activities. *Nuclear Instruments and Methods in Physics Research Section A* 1064, 169681. https://doi.org/10.1016/j.nima.2024.169681

Frankemölle, J.P.K.W., Camps, J., De Meutter, P., Meyers, J., 2025. A Bayesian method for predicting background radiation at environmental monitoring stations in local-scale networks. *Geoscientific Model Development* 18(6), 1989–2003. https://doi.org/10.5194/gmd-18-1989-2025

Hsieh, T.C., Li, C.H., Li, S.H., 2026. A minute-level analysis between environmental gamma dose rate and precipitation. *IEEE Access* 13, 3649414. https://doi.org/10.1109/ACCESS.2025.3649414

ICRP, 1996. Conversion Coefficients for Use in Radiological Protection against External Radiation (ICRP Publication 74). *Annals of the ICRP* 26(3–4), 5–19. https://doi.org/10.1016/S0146-6453(96)90003-2

Karafasoulis, K., Kyriakis, A., 2026. Anomaly detection in gamma-ray spectra using autoencoders with small form factor CZT detectors. *Applied Radiation and Isotopes* 230, 112409. https://doi.org/10.1016/j.apradiso.2025.112409

Ke, G., Meng, Q., Finley, T., Wang, T., Chen, W., Ma, W., Ye, Q., Liu, T.Y., 2017. LightGBM: A highly efficient gradient boosting decision tree. *Advances in Neural Information Processing Systems (NeurIPS 2017)* 30, 3146–3154. https://proceedings.neurips.cc/paper_files/paper/2017/file/6449f44a102fde848669bdd9eb6b76fa-Paper.pdf

Knoll, G.F., 2010. Radiation Detection and Measurement, fourth ed.. *John Wiley & Sons, Hoboken, NJ*. https://www.wiley.com/en-us/Radiation+Detection+and+Measurement%2C+4th+Edition-p-9780470131480

Lalor, P., Adams, H., Hagen, A., 2026. Sim-to-real supervised domain adaptation for radioisotope identification. *Nuclear Instruments and Methods in Physics Research Section A* 1083, 171159. https://doi.org/10.1016/j.nima.2025.171159

Lye, A., Ong, Z.Y., Xiao, N., Patelli, E., 2025. Physics-enhanced machine learning for probabilistic risk assessment in nuclear safety: An overview, recent developments, and perspectives. *Annals of Nuclear Energy* 218, 111562. https://doi.org/10.1016/j.anucene.2025.111562

Masson, O., Baeza, A., Bieringer, J., Brudecki, K., et al., 2011. Tracking of airborne radionuclides from the damaged Fukushima Dai-Ichi nuclear reactors by European networks. *Environmental Science & Technology* 45(18), 7670–7677. https://doi.org/10.1021/es2017158

Melintescu, A., Chambers, S.D., Crawford, J., Williams, A.G., Zorila, B., Galeriu, D., 2018. Radon-222 related influence on ambient gamma dose. *Journal of Environmental Radioactivity* 189, 67–78. https://doi.org/10.1016/j.jenvrad.2018.03.012

Meng, L.J., Ramsden, D., 2000. An inter-comparison of three spectral-deconvolution algorithms for gamma-ray spectroscopy. *IEEE Transactions on Nuclear Science* 47(4), 1329–1336. https://doi.org/10.1109/23.872973

Mercier, J.F., Tracy, B.L., d'Amours, R., Chagnon, F., Hoffman, I., Ungar, R.K., Korpach, E.P., 2009. Increased environmental gamma-ray dose rate during precipitation: a strong correlation with contributing air mass. *Journal of Environmental Radioactivity* 100(7), 527–533. https://doi.org/10.1016/j.jenvrad.2009.03.002

Minato, S., 1980. Analysis of time variations in natural background gamma radiation flux density. *Journal of Nuclear Science and Technology* 17(6), 461–469. https://doi.org/10.1080/18811248.1980.9732610

Moszyński, M., Nassalski, A., Syntfeld-Każuch, A., Szczęśniak, T., Czarnacki, W., Wolski, D., Pausch, G., Stein, J., 2006. Temperature dependences of $LaBr_3$(Ce), $LaCl_3$(Ce) and NaI(Tl) scintillators. *Nuclear Instruments and Methods in Physics Research Section A* 568(2), 739–751. https://doi.org/10.1016/j.nima.2006.06.039

Paatero, J., 2000. Wet deposition of radon-222 progeny in northern Finland measured with an automatic precipitation gamma analyser. *Radiation Protection Dosimetry* 87(4), 273–280. https://doi.org/10.1093/oxfordjournals.rpd.a033008

Sangiorgi, M., Hernández-Ceballos, M.A., Jackson, K., Cinelli, G., Bogucarskis, K., De Felice, L., Patrascu, A., De Cort, M., 2020. The European Radiological Data Exchange Platform (EURDEP): 25 years of monitoring data exchange. *Earth System Science Data* 12(1), 109–118. https://doi.org/10.5194/essd-12-109-2020

Smith, A., Lott, N., Vose, R., 2011. The Integrated Surface Database: Recent developments and partnerships. *Bulletin of the American Meteorological Society* 92(6), 704–708. https://doi.org/10.1175/2011BAMS3015.1

Sportisse, B., 2007. A review of parameterizations for modelling dry deposition and scavenging of radionuclides. *Atmospheric Environment* 41(13), 2683–2698. https://doi.org/10.1016/j.atmosenv.2006.11.057

Stohl, A., Seibert, P., Wotawa, G., Arnold, D., Burkhart, J.F., Eckhardt, S., Tapia, C., Vargas, A., Yasunari, T.J., 2012. Xenon-133 and caesium-137 releases into the atmosphere from the Fukushima Dai-ichi nuclear power plant: determination of the source term, atmospheric dispersion, and deposition. *Atmospheric Chemistry and Physics* 12(5), 2313–2343. https://doi.org/10.5194/acp-12-2313-2012

Sun, J., Niu, D., Liang, J., Li, L., 2024. Rapid nuclide identification algorithm based on self-attention mechanism neural network. *Annals of Nuclear Energy* 207, 110708. https://doi.org/10.1016/j.anucene.2024.110708

Takeuchi, N., Katase, A., 1982. Rainout-washout model for variation of environmental gamma-ray intensity by precipitation. *Journal of Nuclear Science and Technology* 19(5), 393–409. https://doi.org/10.1080/18811248.1982.9734160

Wen, S., Zhou, R., 2024. Temperature peak-drift correction method for NaI(Tl) detectors using the background peak. *Sensors* 24(8), 2621. https://doi.org/10.3390/s24082621